

Mandatory Exercise 02282 F26

Admin

Follow the guidelines on the homepage on how to submit your solution. Remember to use the template and to read the collaboration policy. The page limit is one page.

Placement of distribution centers

You are consulting for a transport company that distributes goods from their distribution centers. They are going to start up in a new area, and want to know where to place their distribution centers. The map of the area is given as a graph $G = (V, E)$ with n nodes and distance function d , which is a metric. Some places have higher priority than others. The company has partitioned the potential customers (vertices) into two groups (pink and green). They want to place k centers, such that they minimize the maximum *prioritized* distance of a vertex to its closest center: If v is pink then the prioritized distance $d_p(v, c)$ from v to a center c is $2 \cdot d(v, c)$. If v is green then the prioritized distance $d_p(v, c)$ from v to a center c is $d(v, c)$.

Formally, the problem is to find a set of centers $C \subseteq V$ with $|C| \leq k$ minimizing

$$r(C) = \max_{v \in V} d_p(v, C),$$

where

$$d_p(v, C) = \min_{u \in C} d_p(v, u).$$

1 Approximation algorithm We want a 2-approximation algorithm for the problem.

Note: You may assume you know the optimum covering radius.

- 1.1 Describe your algorithm.
- 1.2 Argue that your algorithm outputs at most k centers.
- 1.3 Argue that the approximation factor of your algorithm is 2.
- 1.4 Argue that your algorithm runs in polynomial time.